

Publish Hugo with GitHub Actions

Push source changes; let the workflow build and publish the site.

Learning Goals

- Push Hugo source and publish the site automatically.
- Explain the update loop: edit, push, and GitHub builds and publishes for you.
- Find the first failed step when a deployment fails.

The Key Automation: Push = Deploy

This is the most important automation in Module 3.

```
edit → commit + push → GitHub Actions builds Hugo → your site updates
```

After the one-time setup, updating your website needs only a push.

	Manual (lesson 1)	Automatic (this lesson)
Build the site	you run <code>./run.sh</code>	GitHub runs <code>hugo</code>
Copy files	you run <code>cp</code>	none
Publish	commit and push	commit and push

You never build or copy again. Edit your files, push, and check the **Actions** tab.

1. Prepare Your Site Repository

Create or clone **your own** `yourusername.github.io` repository (see `githubio_installation.pdf`).

- Copy the contents of `my-portfolio/` to its root:
- Do not copy `.git/` or the generated `public/` .

```
content/    layouts/    static/  
archetypes/ hugo.toml
```

2. Copy the Workflow

The supplied files are under:

```
module3/code/3_Github_io/3_Github_io_Hugo_Deployment  
/.
```

- Copy `workflows/hugo.yml` to `.github/workflows/hugo.yml`.
- Add the entries from the supplied `gitignore` to the `.gitignore` at your site repository root.
- If you have none, save it as `.gitignore`.
- Keep your own rules; the required ones are `public/`, `resources/_gen/`, `.hugo_build.lock`, and the `.env` rules that keep secrets out of a public repository.
- The workflow installs the Hugo version set by `HUGO_VERSION` at the top of `hugo.yml`.

Prepare Files with `setup.sh`

At the time you run the setup script, all three directories are siblings:

```
├── 3_Github_io_Hugo_Deployment
│   ├── gitignore
│   ├── setup.sh
│   └── workflows
│       └── hugo.yml
├── my-portfolio
└── yourusername.github.io
```

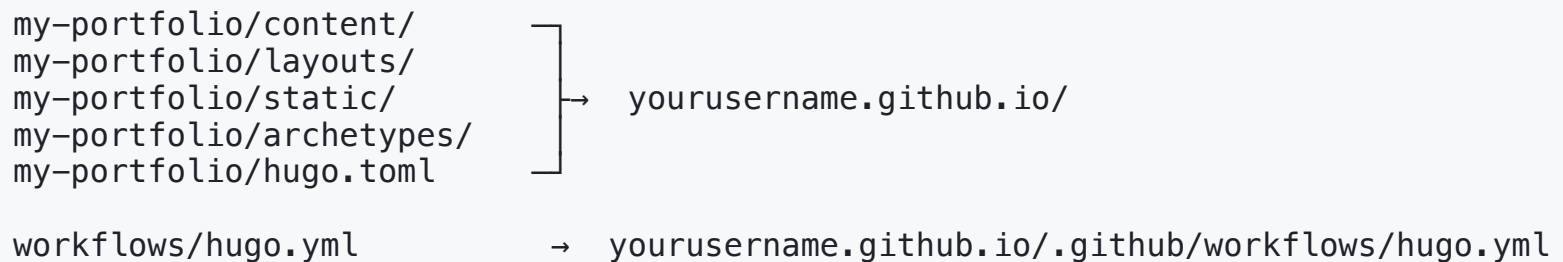
`setup.sh` is inside `3_Github_io_Hugo_Deployment/`, next to the supplied `gitignore` and `workflows/` directory.

- The third directory is the Pages repository that you cloned in Step 1. Move into the deployment directory and run:

```
cd 3_Github_io_Hugo_Deployment
./setup.sh ../yourusername.github.io
```

What `setup.sh` Copies

It copies two things into your Pages repository:



The diagram illustrates the mapping of local files to a GitHub repository. On the left, a list of local files is shown: `my-portfolio/content/`, `my-portfolio/layouts/`, `my-portfolio/static/`, `my-portfolio/archetypes/`, and `my-portfolio/hugo.toml`. A right-facing curly bracket groups these five items. An arrow points from this bracket to the text `yourusername.github.io/`. Below this, the file `workflows/hugo.yml` is listed, with an arrow pointing to `yourusername.github.io/.github/workflows/hugo.yml`.

```
my-portfolio/content/  
my-portfolio/layouts/  
my-portfolio/static/  
my-portfolio/archetypes/  
my-portfolio/hugo.toml  
  
workflows/hugo.yml
```

→ yourusername.github.io/

→ yourusername.github.io/.github/workflows/hugo.yml

It also adds the supplied rules to your `.gitignore`.

- Run it **only once**. The repository `yourusername.github.io/` is now your Hugo source.
- From now on, edit in `yourusername.github.io/` (for example `content/`), preview with `hugo server -D` , and push.
Nothing copies `my-portfolio/` into it again, so stop editing `my-portfolio/` .
- It never commits, pushes, or deletes files. Running it again stops with `Refusing to overwrite existing path` .

3. Configure Pages

- On GitHub, open **Settings** → **Pages**. Set **Source** to **GitHub Actions**.
- In the manual lesson, the source was **Deploy from a branch**. Change it now.

GitHub Pages

GitHub Pages is designed to host your personal, organization, or project pages

Your site is live at <https://nkuase.github.io/>
Last **deployed** by  **nkuase** 5 minutes ago

Build and deployment

Source

GitHub Actions ▾

4. Push from VS Code

Open your `yourusername.github.io` folder in VS Code, then commit and push (**Sync Changes**).

- Commit and sign-in steps: `githubio_installation.pdf`.

```
VS Code push → GitHub runs Hugo (compile) → deploy → https://yourusername.github.io/
```

That is all. GitHub compiles Hugo and publishes the site for you.

5. Verify Publication

- Open **Actions** and inspect the build and deploy jobs.
 - Open `https://yourusername.github.io/`.
 - Check the About page, API documentation, screenshots, and links.
 - Change a sentence in `yourusername.github.io/content/`, push again, and confirm the live page updates.
- A push is all it takes; GitHub builds and publishes it for you.